



Agenda



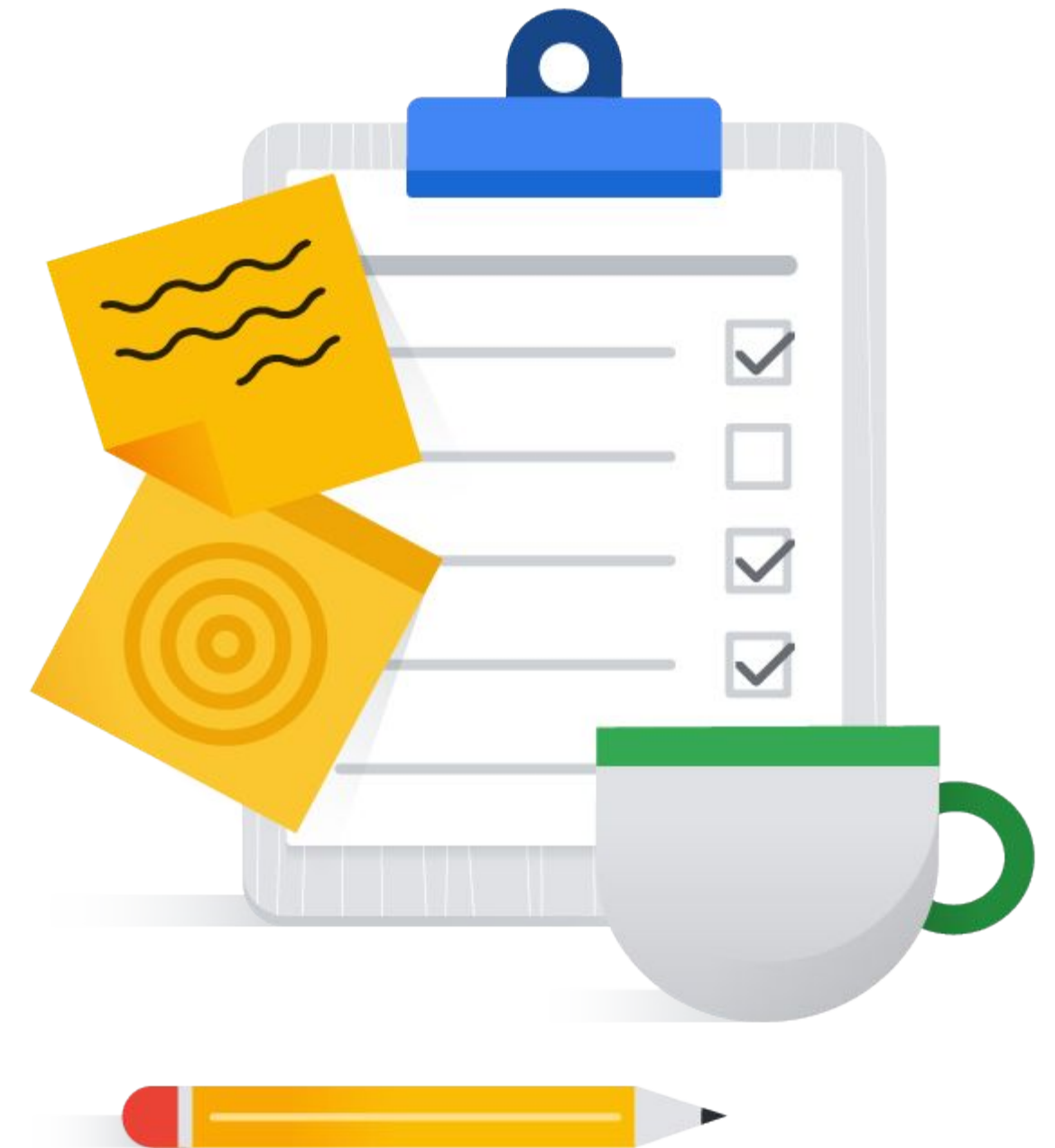
- 01 Introducing Agent Development Kit
- 02 Develop agents with ADK
- 03 Build multi-agent systems with ADK
- 04 Deploy ADK agents to Agent Engine
- 05 Evaluate ADK agent systems



Build multi-agent systems with ADK

Module objectives

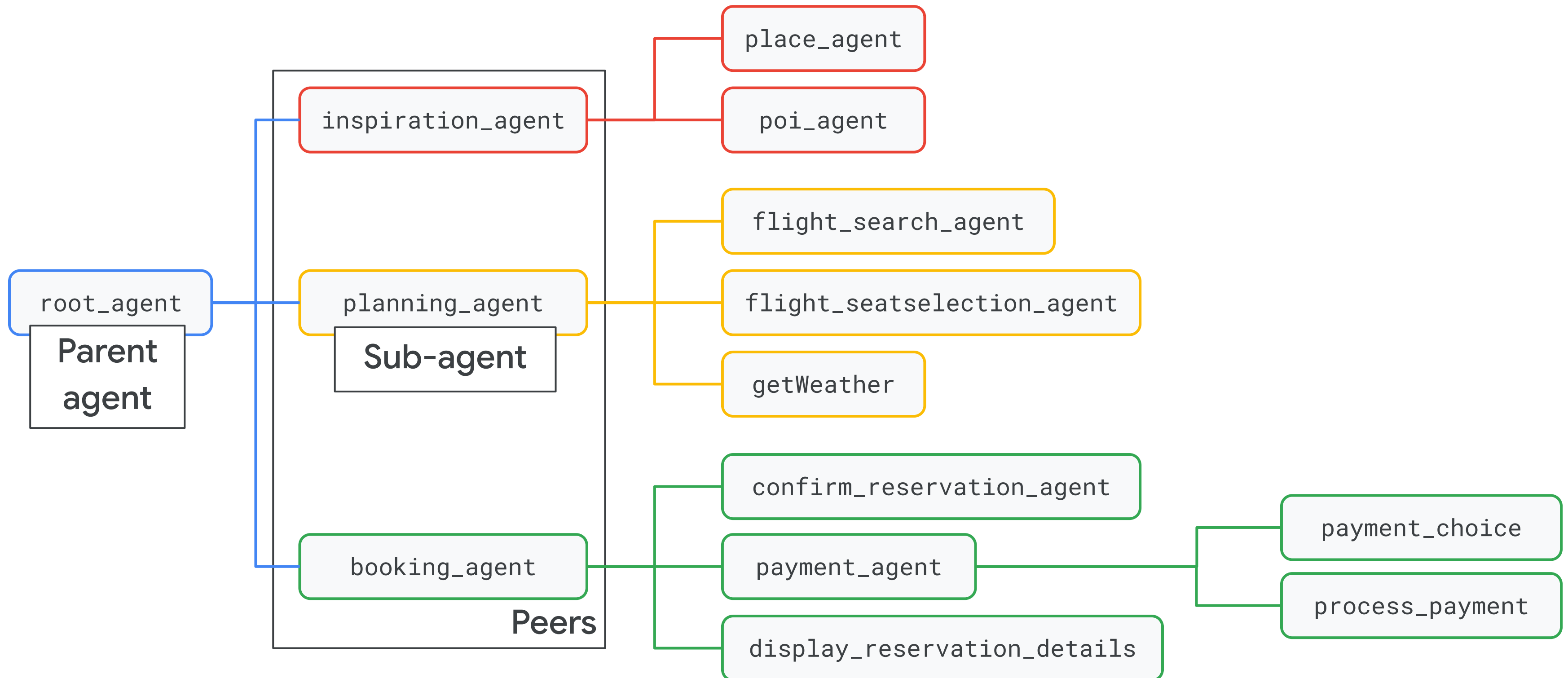
01	Describe the concept of multi-agent systems in ADK.
02	Differentiate between LLM-based and workflow agents.
03	Explain the purpose of different types of workflow agents.
04	Recognize when to use a custom workflow agent.



Introducing multi-agent systems



A hierarchical approach to multi-agent systems



Agents forming part of multi-agent systems



Large Language Model-based agents

- ✓ Bring intelligence and adaptability to your system
- ✓ Understand and respond to natural language
- ✓ Make decisions
- ✓ Utilize tools effectively



Workflow agents

- ✓ Create flows directly from agent to agent
- ✓ Act as directors, to ensure tasks are performed in sequence and under specific conditions
- ✓ Control the flow of the context between their sub-agents
- ✓ Don't have LLMs associated - they don't reason themselves
- ✓ Deterministic - for a given input and configuration, the sequence of agents executed will always be the same



Types of workflow agents

SequentialAgent

LoopAgent

ParallelAgent

Custom



Types of workflow agents

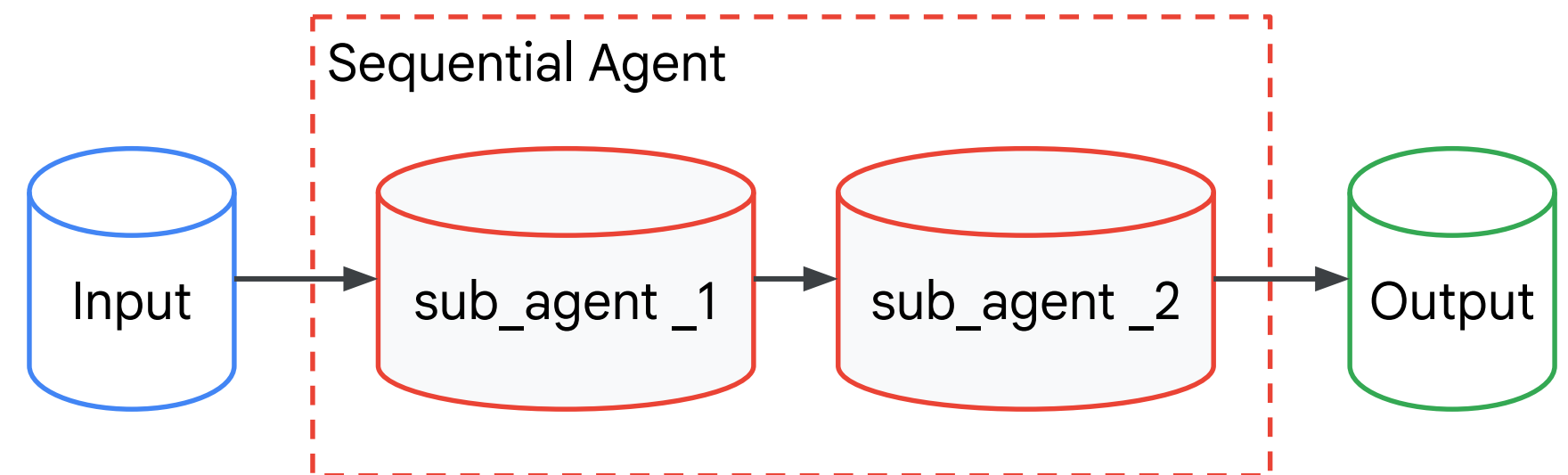
SequentialAgent

LoopAgent

ParallelAgent

Custom

- Executes agents in a specific order
- For workflows with a fixed order, when tasks have dependencies requiring sequential execution



Types of workflow agents

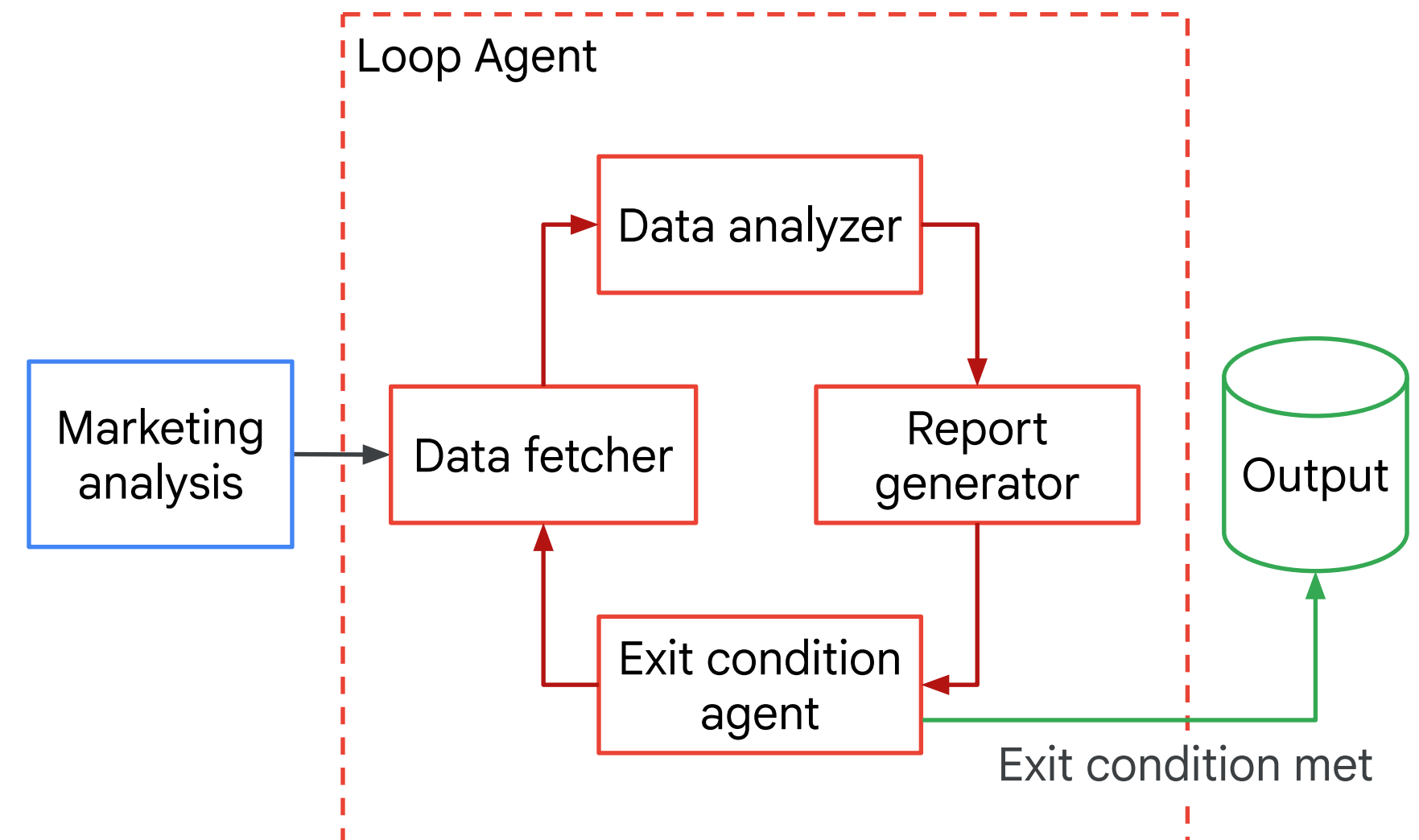
SequentialAgent

LoopAgent

ParallelAgent

Custom

- Executes agents repeatedly, until a certain condition is met
- For tasks that require iteration, continuous monitoring, cyclical processes or simulated negotiation



Types of workflow agents

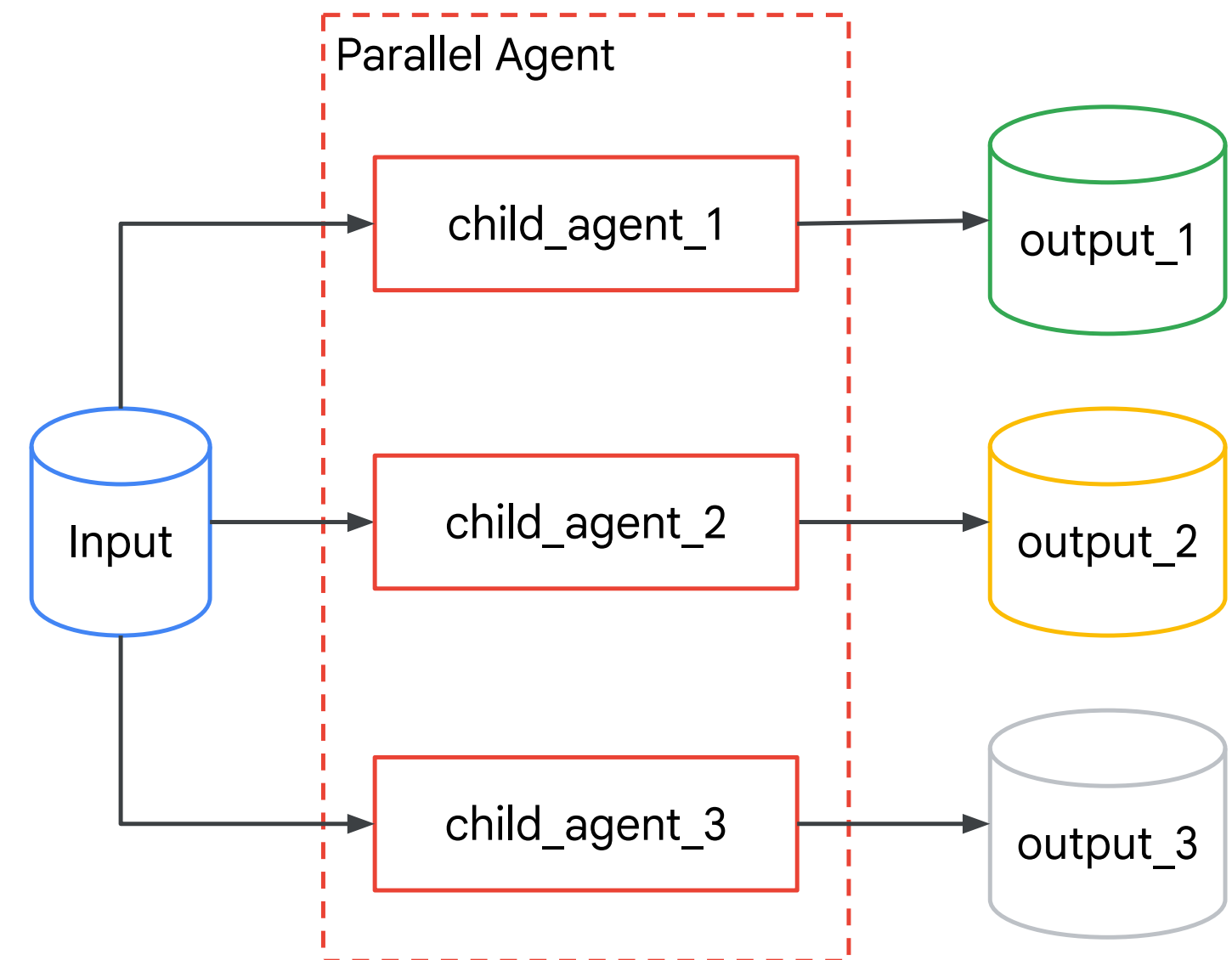
SequentialAgent

LoopAgent

ParallelAgent

Custom

- Executes agents simultaneously
- For tasks with independent sub-tasks, tasks that prioritize speed, or are resource intensive



Types of workflow agents

SequentialAgent

LoopAgent

ParallelAgent

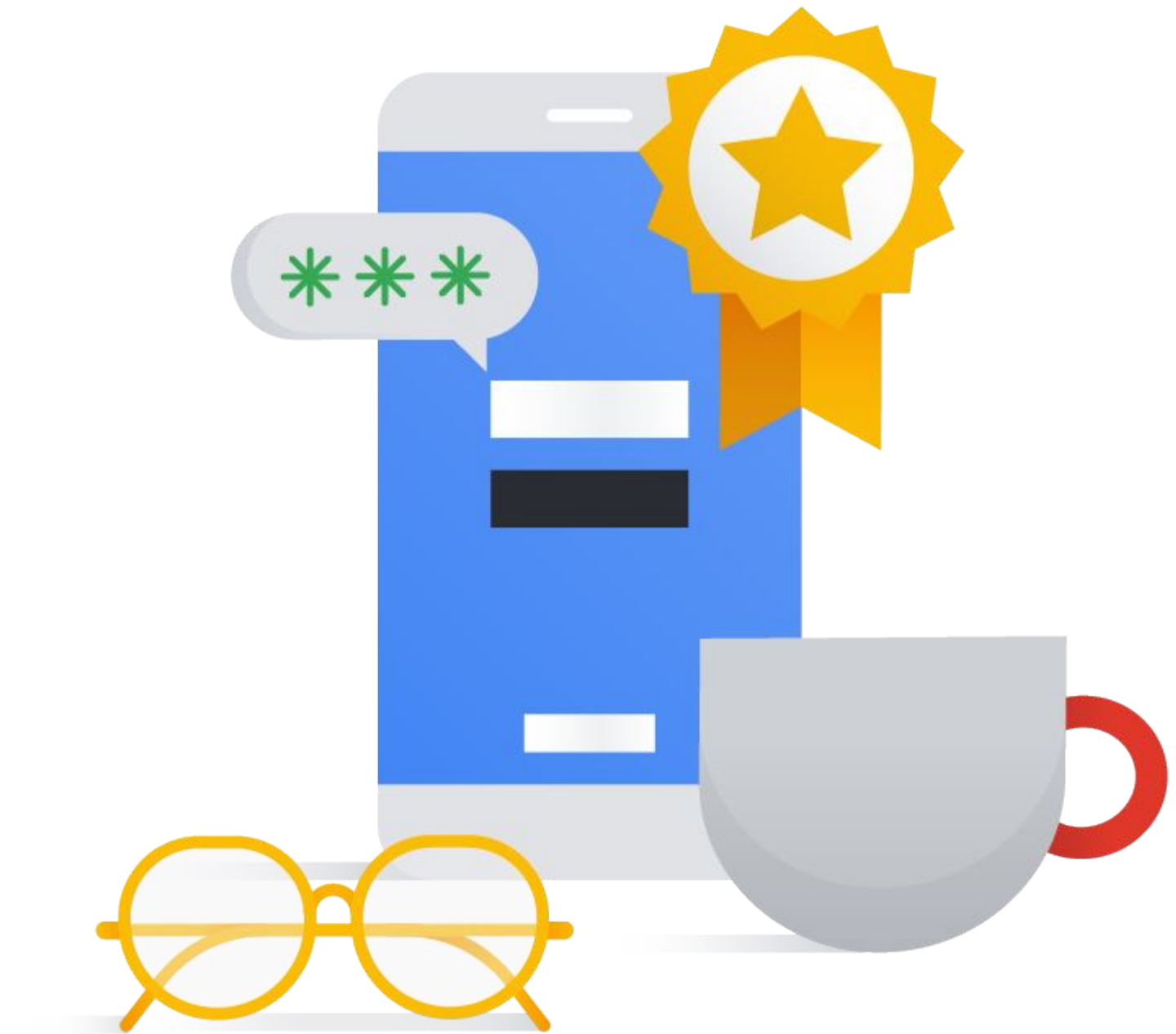
Custom

Use when your requirements include:

- Conditional logic
- Complex state management
- External integrations
- Dynamic agent selection
- Unique workflow patterns



Now let's do a short
quiz to **check your
knowledge!**



Quiz | Question 01

Question

What is the key idea behind the hierarchical approach to multi-agent systems?

- A. Agents can only transfer to sub-agents, parents, or peers
- B. Agents can transfer to any other agent
- C. Agents can only transfer to sub-agents or back to a parent
- D. Agents can only transfer to parent or peer agents

Quiz | Question 01

Answer

What is the key idea behind the hierarchical approach to multi-agent systems?

- A. Agents can only transfer to sub-agents, parents, or peers.
- B. Agents can transfer to any other agent.
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- D. Agents can only transfer to parent or peer agents.



Quiz | Question 02

Question

What is the primary function of workflow agents?

- A. To process natural language with LLMs.
- B. To act as "brains" of the application.
- C. To execute tasks independently of each other.
- D. To create flows directly from agent to agent.

Quiz | Question 02

Answer

What is the primary function of workflow agents?

- A. To process natural language with LLMs.
- B. To act as "brains" of the application.
- C. To execute tasks independently of each other.
- D. To create flows directly from agent to agent.



Quiz | Question 03

Question

What is a key benefit of using ParallelAgents?

- A. They execute agents in a specific order.
- B. They execute agents repeatedly until a condition is met.
- C. They execute agents simultaneously for speed.
- D. They manage workflows for complex business logic.

Quiz | Question 03

Answer

What is a key benefit of using ParallelAgents?

- A. They execute agents in a specific order.
- B. They execute agents repeatedly until a condition is met.
- C. They execute agents simultaneously for speed.
- D. They manage workflows for complex business logic.

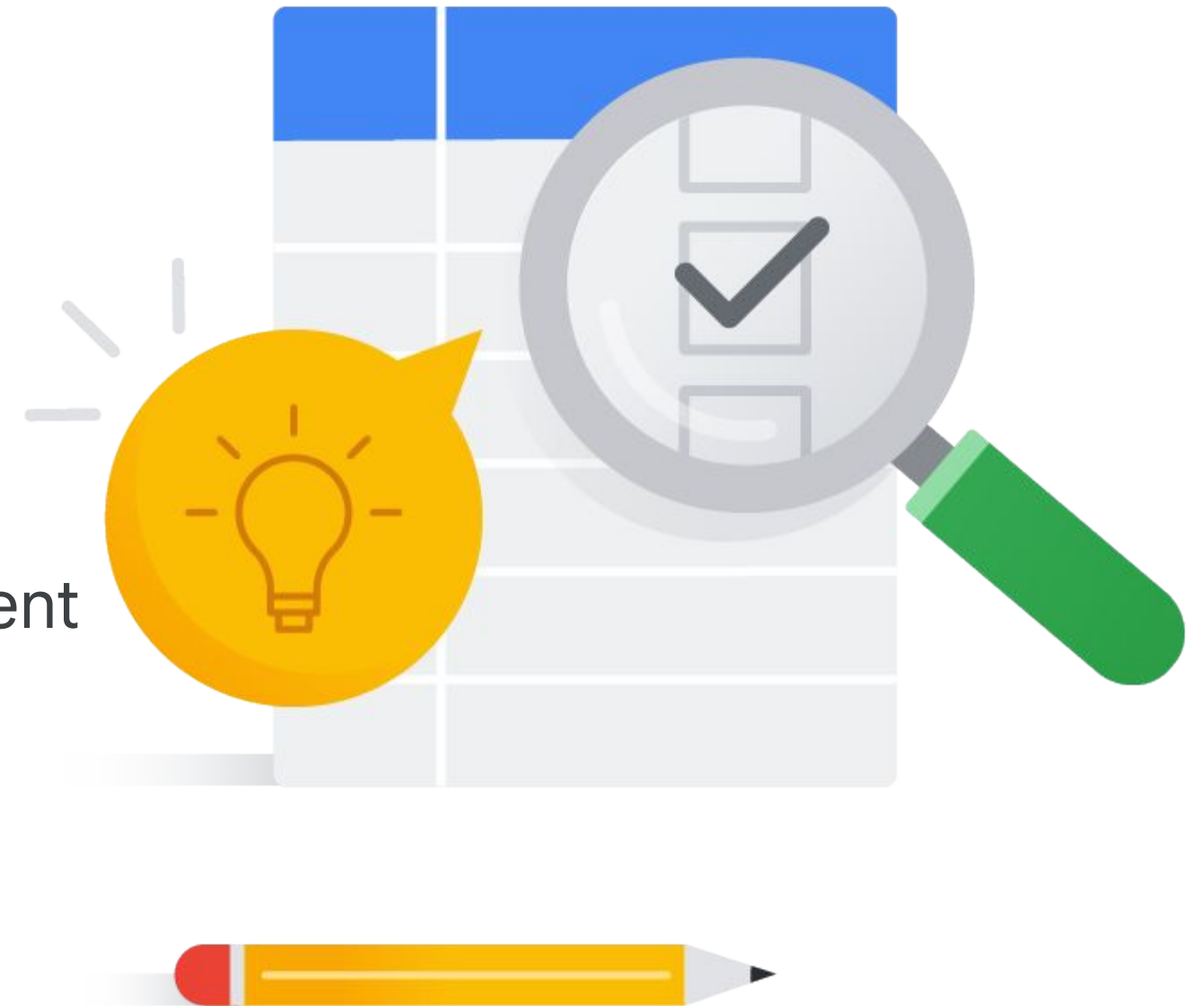


Lab: Build multi-agent systems with ADK

🕒 45 min

Overview

In this lab, you will learn about multi-agent systems using Agent Development Kit.



Google Cloud